Database Technologies – Assignment 10 PL-SQL

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = 2*(L+W) and Area = L*W.

```
D2_96583_rina@localhost DELIMITER $$
D2_96583_rina@localhost CREATE PROCEDURE p_rectang(p_length INT, p_breadth I
NT)
    -> BEGIN
    -> DECLARE v_res1 INT;
    -> DECLARE v_res2 INT;
    -> SELECT p_length * p_breadth INTO v_res1;
    -> SELECT 2*(p_length + p_breadth) INTO v_res2;
    -> INSERT INTO RESULTS VALUES(v_res1, 'area_of_rectangle');
    -> INSERT INTO RESULTS VALUES(v_res2, 'perimeter_of_rectangle');
    -> END;
    -> $$
Query OK, 0 rows affected (0.074 sec)

D2_96583_rina@localhost DELIMITER ;
D2_96583_rina@localhost TRUNCATE RESULTS;
Query OK, 0 rows affected (0.194 sec)

D2_96583_rina@localhost CALL p_rectang(10, 10);
Query OK, 1 row affected (0.103 sec)

D2_96583_rina@localhost SELECT * FROM results;
+------+------------------------+
| v1   | v2                     |
+------+------------------------+
|  100 | area_of_rectangle      |
|   40 | perimeter_of_rectangle |
+------+------------------------+
2 rows in set (0.008 sec)

D2_96583_rina@localhost
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
D2_96583_rina@localhost CREATE TABLE TEMPORARY(original_vale INT, val_cube I
NT, val_square INT);
Query OK, 0 rows affected (0.168 sec)

D2_96583_rina@localhost DELIMITER $$
D2_96583_rina@localhost CREATE PROCEDURE CALCULATE()
    -> BEGIN
    -> DECLARE num INT;
    -> SET num = 10;
    -> INSERT INTO TEMPORARY VALUES(num, pow(num, 3), pow(num, 2));
    -> END;
    -> $$
Query OK, 0 rows affected (0.157 sec)

D2_96583_rina@localhost DELIMITER ;
D2_96583_rina@localhost CALL CALCULATE();
Query OK, 1 row affected (0.088 sec)

D2_96583_rina@localhost SELECT * FROM TEMPORARY;
+---------------+----------+------------+
| original_vale | val_cube | val_square |
+---------------+----------+------------+
|            10 |     1000 |        100 |
+---------------+----------+------------+
1 row in set (0.008 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are C= (F-32) *5/9 F= 9/5*C + 32

```
D2_96583_rina@localhost DELIMITER $$
D2_96583_rina@localhost CREATE PROCEDURE ConvertTemperature2(IN input_temp D
ECIMAL(10,2), IN unit CHAR(1))
    -> BEGIN
    -> DECLARE res2 DECIMAL(10, 2);
    ->  IF UPPER(unit) = 'C' THEN
    -> SET res2 = (input_temp - 32) *(5/9);
    -> ELSEIF UPPER(unit) = 'F' THEN
    -> SET res2 = (input_temp * 9 / 5) + 32;
    -> ELSE
    -> SET res2 = 'INVALI CONVERSION';
    -> END IF;
    -> SELECT res2 AS RESULT;
    -> END
    -> $$
Query OK, 0 rows affected (1.006 sec)

D2_96583_rina@localhost DELIMITER ;
D2_96583_rina@localhost CALL ConvertTemperature2(45, 'F');
+--------+
| RESULT |
+--------+
| 113.00 |
+--------+
1 row in set (0.854 sec)

Query OK, 0 rows affected (0.865 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals yards, 1 foot, and 4 inches.

```
02_96583_rina>DELIMITER $$
02_96583_rina>CREATE PROCEDURE ConvertInchesToUnits(
    ->     IN total_inches INT,
    ->     OUT yards INT,
    ->     OUT feet INT,
    ->     OUT inches INT
    -> )
    -> BEGIN
    ->     -- Calculate whole yards
    ->     SET yards = total_inches / 36;
    ->
    ->     -- Calculate remaining feet from what's left
    ->     SET feet = (total_inches % 36) / 12;
    ->
    ->     -- Calculate remaining inches
    ->     SET inches = total_inches % 12;
    -> END;
    -> $$
Query OK, 0 rows affected (1.196 sec)

02_96583_rina>DELIMITER ;
02_96583_rina>CALL ConvertInchesToUnits(124, @y, @f, @i);
Query OK, 0 rows affected (0.322 sec)

02_96583_rina>SELECT @y AS Yards, @f AS Feet, @i AS Inches;
+-------+------+--------+
| Yards | Feet | Inches |
+-------+------+--------+
|     3 |    1 |      4 |
+-------+------+--------+
1 row in set (0.011 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
IF Pro' at line 1
D2_96583_rina@localhost DELIMITER $$
D2_96583_rina@localhost CREATE PROCEDURE p_product(IN p_num1 INT, IN p_num2
INT)
    -> BEGIN
    -> DECLARE product INT;
    -> SET product = p_num1 * p_num2;
    -> IF Product >= 100 THEN
    -> SELECT product AS "PRODUCT", 'EQUAL TO OR GREATER  THAN 100' AS STATU
S;
    -> ELSE
    -> SELECT product AS "PRODUCT", 'LESS THAN 100' AS STATUS;
    -> END IF;
    -> END
    -> $$
Query OK, 0 rows affected (0.906 sec)

D2_96583_rina@localhost CALL p_product(10, 12);
    -> ^C
D2_96583_rina@localhost delimiter ;
D2_96583_rina@localhost CALL p_product(10, 12);
+---------+------------------------------+
| PRODUCT | STATUS                       |
+---------+------------------------------+
|     120 | EQUAL TO OR GREATER  THAN 100 |
+---------+------------------------------+
1 row in set (0.047 sec)

Query OK, 0 rows affected (0.061 sec)

D2_96583_rina@localhost |
```